Presentation and challenge session

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Final project manuscript. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of presentation and challenge session. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of presentation and **challenge session**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Final project manuscript. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of presentation and **challenge session**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Final project manuscript. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain presentation and **challenge session** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Presentation and challenge session case. Separate observations, given facts and assumptions before choosing an explanation.

A presentation should make the **evidence** easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Presentation and challenge session. Connect each labelled input, deciding step and result.

Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating **evidence** or stating it is missing.

Presentation and challenge session — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A presenter shows one perfect output after a20-case study with14 passes.	Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.	The demonstration is one example; study rate70%. Present aggregate and failures and do not call the best output representative without evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Presentation and challenge session. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A presenter shows one perfect **output** after a20-case study with14 passes. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: The reviewer shows one camera-B success.

Predicted result

Prediction: That is one demonstration, not an estimate of broad camera-B reliability.

Why the result follows

Reason: One selected **observation** does not represent the full proposed condition.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Presentation and challenge session to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Challenge session	Questions testing the basis of a presentation
Representative example	An example whose selection is justified for the stated purpose
Evidence response	An answer grounded in records or explicit absence

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Presentation and challenge session, predict a step, then compare with its explanation.

A presenter shows one perfect **output** after a20-case study with14 passes.

The demonstration is one example; study rate**70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Worked explanation · Presentation and challenge session

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Presentation and challenge session

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

The facts we will use

Supplied facts

The project tested twelve photos from camera A only. Ten were correct; no camera B records exist.

AI ACADEMY / AI-7/35

Presentation and challenge session

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State measured result

Report 10/12 correct on camera A.



2. Address the challenge

Say camera B performance is not measured.



3. Propose a valid next test

Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

CHANGE ONE THING

The reviewer shows one camera-B success.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State measured result	Report 10/12 correct on camera A.
2. Address the challenge	Say camera B performance is not measured.
3. Propose a valid next test	Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

What this example does not establish

Keep the stated limit

The original result and proposed study still need sampling and uncertainty discussion.

Keep the stated limit

Read the labelled picture: Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

Keep the stated limit

Change one thing: The reviewer shows one camera-B success. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Presentation and challenge session claim and explain a repair using the supplied evidence.

A confident response should not replace missing **evidence** with speculation.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A confident response should not replace missing **evidence** with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Presentation and challenge session with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Presentation and **challenge session**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Presentation and **challenge session**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Presentation and challenge session. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A presenter shows one perfect **output** after a20-case study with14 passes. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Presentation and challenge session. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A confident response should not replace missing **evidence** with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Presentation and challenge session case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Presentation and challenge session — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A presenter shows one perfect output after a20-case study with14 passes.	Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.	The demonstration is one example; study rate70%. Present aggregate and failures and do not call the best output representative without evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Presentation and challenge session, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

By the end, I can	How I will show it
Explain the mechanism and terminology of presentation and challenge session	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Presentation and challenge session without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of presentation and **challenge session**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

1. Propose a valid next test

2. State measured result

3. Address the challenge

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Propose a valid next test Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The reviewer shows one camera-B success. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That is one demonstration, not an estimate of broad camera-B reliability.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

Repair a missing building block

Review Final project manuscript (ai-7/34). Goal: Explain the **mechanism** and terminology of final project manuscript. Short practice: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping.

Which abstract claim must be corrected if it says ten independent plants?. Return to this week and explain what you changed.

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Presentation and challenge session graphic using its labels, deciding evidence and relevant limitation.

01. Presentation and challenge session: the case



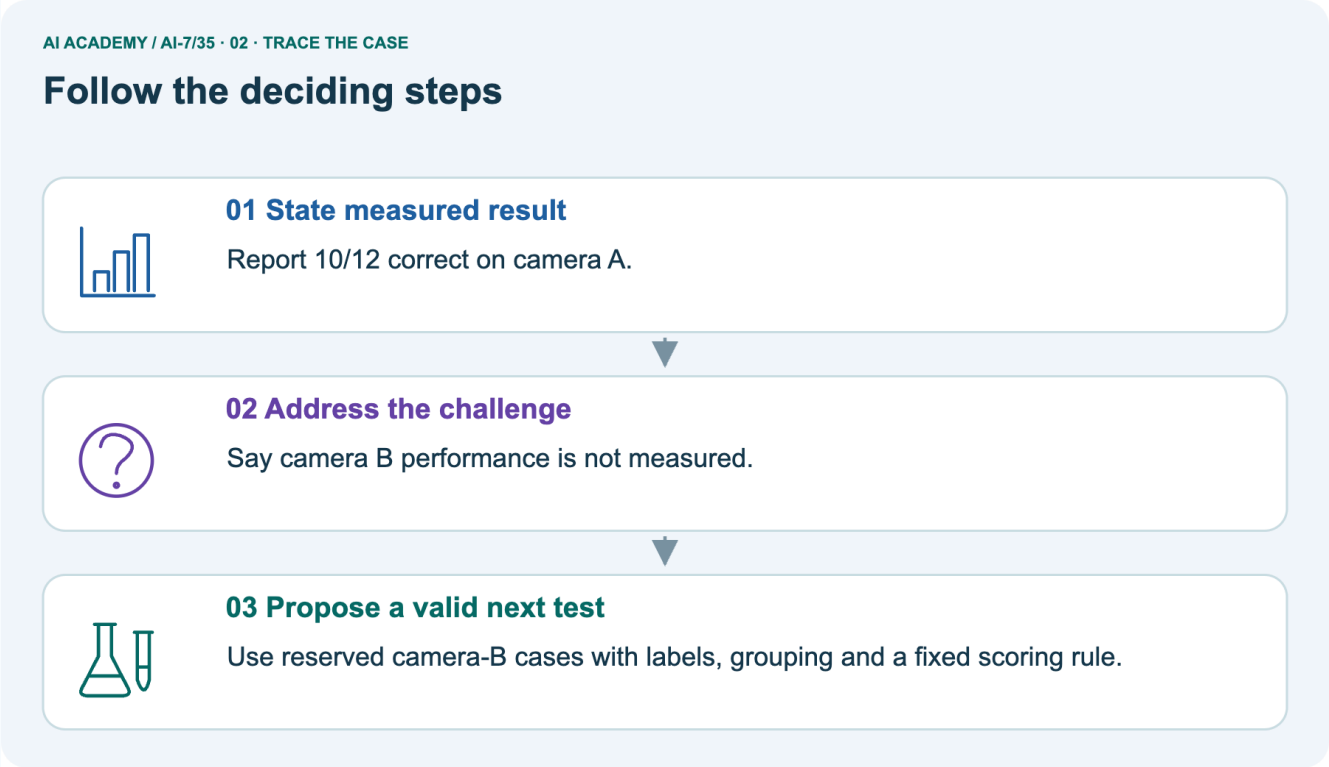
Follow the labelled picture with the matching explanation. Open the vector diagram

Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera. Supplied facts: The project tested twelve photos from camera A only.

Ten were correct; no camera B records exist.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

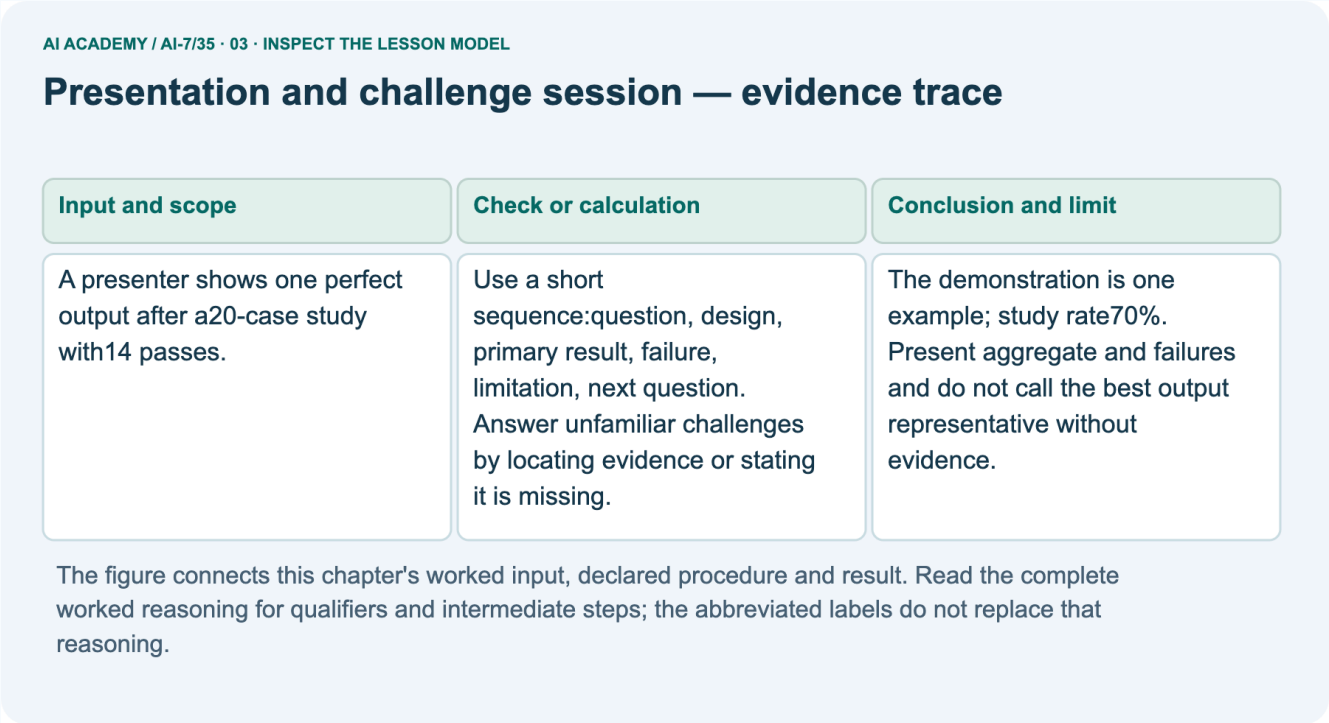


Follow the labelled picture with the matching explanation. Open the vector diagram

State measured result: Report 10/12 correct on camera A. → Address the challenge: Say camera B performance is not measured. → Propose a valid next test: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Point to each numbered stage and explain which recorded **input** it uses.

03. Presentation and challenge session — evidence trace



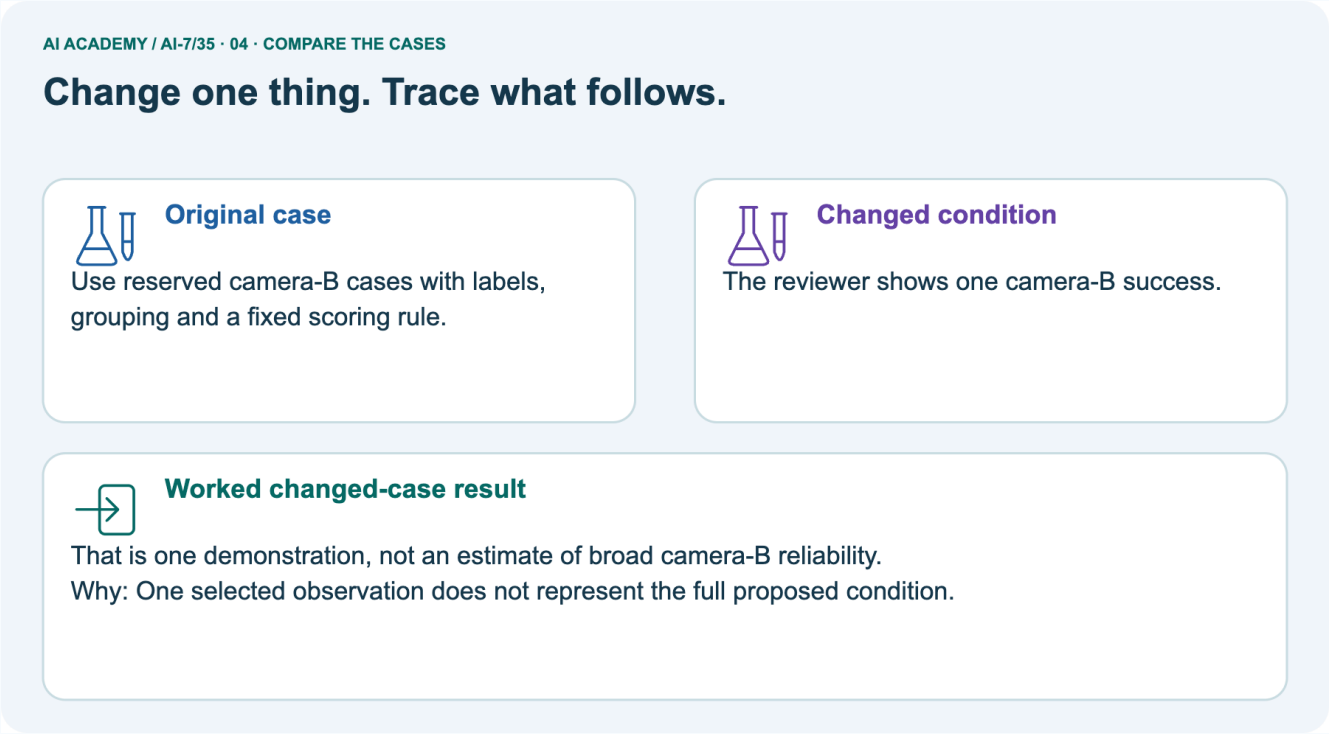
Follow the labelled picture with the matching explanation. Open the vector diagram

Presentation and **challenge session** — **evidence** trace. A presenter shows one perfect **output** after a20-case study with14 passes. | Use a short sequence:question, design, primary result, failure, limitation, next question.

Answer unfamiliar challenges by locating **evidence** or stating it is missing. | The demonstration is one example; study rate**70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



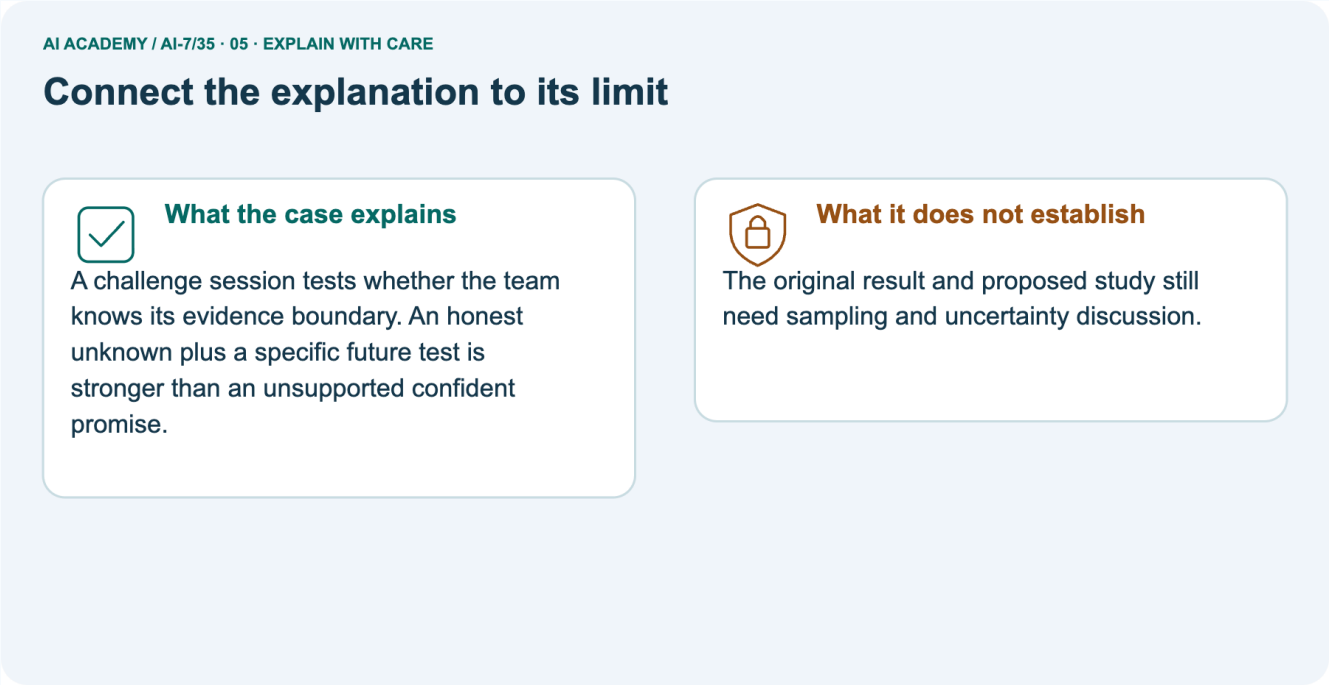
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.
Changed condition: The reviewer shows one camera-B success.

Expected result: That is one demonstration, not an estimate of broad camera-B reliability. Reason: One selected **observation** does not represent the full proposed condition.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



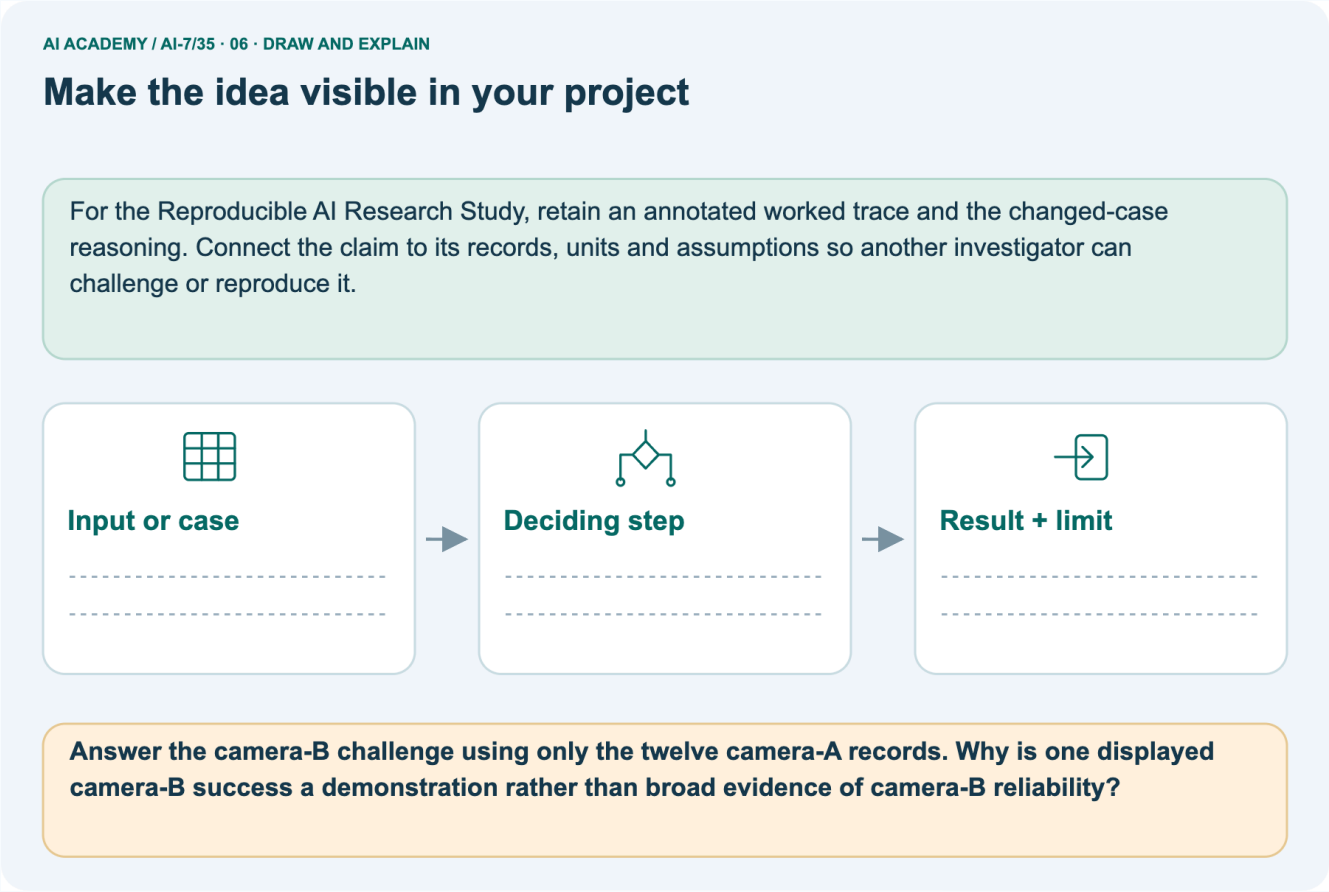
Follow the labelled picture with the matching explanation. Open the vector diagram

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Limit: The original result and proposed study still need sampling and uncertainty discussion.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.